Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/29/2024
PI Name:	Wood, Scott J. Ph.D.		
Project Title:	Manual Crew Override of Vehicle Landings Following G-Transitions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI returned to NASA JSC in January 2017. PI was at Azusa Pacific University from August 2013 – January 2017; prior to August 2013, PI was at NASA JSC.		
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/04/2022	**End Date:**	09/30/2032
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Steven Moore and Raymond Heineman have been removed due to changes in the experiment protocol, including removal of the inflight piloting task (T-38 and Pilatus PC-12) and changes to the test battery. Michael Bishop resigned from his KBR position.		
COI Name (Institution):	Duda, Kevin Ph.D. (Draper Laboratory) Wheelock, Douglas M.S. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Manual control during exploration spaceflight consists of both planned automated supervisory control and unplanned crew override. This crew override capability is critical to enable overall mission success during landing contingencies. However, the introduction of manual override capabilities must be implemented to enable crews to mitigate risks introduced by human error. Adaptive changes in the sensorimotor system can manifest during G-transitions as spatial disorientation. While training and landing aids enable successful landing through disorientation, these adaptive changes may increase cognitive demand that need to be accounted for in the manual control strategy. It is important to characterize these effects as soon as possible following the G-transition. The primary goals of this study are (1) to understand the impact of spaceflight on crew ability to perform manual crew override tasks including the effect of flight duration, (2) to examine how adaptive changes in vestibular and cognitive function relate to changes in manual crew override proficiency, and (3) compare performance during late in-flight “just-in-time” on-board training with early post-flight crew performance. This study will recruit astronauts assigned to either short duration Private Astronaut Missions (PAM, < 30 day) or long duration (6-month) missions to the International Space Station (ISS). Ground matched-control subjects tested with the same schedule will enable us to assess the effects of learning and/or recency independent of spaceflight. Our first aim is to characterize the impact of spaceflight on crew capability to perform automated supervisory control and manual landing tasks involving crew override following both short (PAM) and longer duration ISS missions. The impact of spaceflight on piloting capability will be assessed from pre- versus post-flight changes in a simulated lunar landing during which crewmembers will manually takeover attitude and rate-of-descent to the nominal or re-designated landing aimpoint during the approach phase. This simulation will be performed on portable fixed base soon after return from ISS at the rally airport and then on R+1 using a 6DOF motion base at JSC. Subjects will also perform an automated supervisory control on a portable tablet. During this task, subjects will monitor simulation of an out-the-window display during lunar approach to determine if they need to execute a divert to an alternate landing site and use a touch screen to select the site that is within the desired landing zone and free of obstacles. The outcome measures include the percent time maintaining actual vehicle states, e.g., attitude and rate-of-descent, within recommended guidance during the landing approach, number and maximum deviation outside limits, time to execute divert to alternate landing site (if needed), error rates. The cognitive performance measures during the simulation will focus on multi-tasking in which crewmembers will respond to display prompts as a secondary task, as well as eye-head tracking to characterize visual reaction times and areas of visual attention during the simulation. We hypothesize that there will be post-flight increases in error rates and time to execute corrective actions. Our second aim is to examine how adaptive changes in vestibular and cognitive function relate to proficiency during supervisory manual control and crew override. The sensorimotor and cognitive test battery is based on a previous manual control study by Moore et al (2019) in which astronauts exhibited significant changes in motion perception, dual-tasking and sleepiness. The dual-tasking has been embedded in both tablet and piloting tasks. Motion perception accuracy and precision has been integrated with motion-base sim at JSC. Motion sickness & sleepiness will be obtained at all timepoints. We hypothesize that that a higher severity of vestibular alterations will be associated with increased flight increases in error rates and time to execute corrective actions. Given that ‘just-in-time” on-board training is an operational expectation for the extended exploration missions, all participants will perform late inflight JIT training. Crew proficiency will be captured inflight during on-board training that will be implemented on a laptop with hand controllers to allow the crewmember to practice the landing task procedures, like the approach implemented for on-board training with Shuttle landing and ISS telerobotic tasks. The metrics captured during the on-board lunar landing simulations will match those of the pre- and postflight simulations. We hypothesize that that proficiency on the “just-in-time” laptop trainer late in mission will be positively correlated with early post-flight proficiency on the same task. This project will deliver an operational demonstration of automated supervisory control and crew override capability following spaceflight and identify potential deficits that may require remediation. Comparison of individual vestibular and cognitive changes with crew override performance will help better characterize the manual control risks associated with sensorimotor alterations. Comparing performance parameters from the on-board training to post-flight performance will help demonstrate transfer of inflight on-board training to post-flight manual crew override performance. The inclusion of “just-in-time” on-board training will ensure we are characterizing changes in override proficiency with this expected countermeasure in place.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This flight study addresses the sensorimotor research emphasis stated in the Human Research Program (HRP) Integrated Research Plan titled “Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks”. One gap associated with this risk (SM-102) is to “characterize the effects of short and long-duration weightlessness on manual control after G-transitions.” This research gap led to the solicitation of a manual control study conducted before and after long duration flights on the ISS to map changes in sensorimotor function to manual control decrements. Unfortunately, these results (Moore et al., 2019) were limited due to testing delays related to time required for direct returns from Kazakhstan, with the initial measurements conducted more than 20 hr following landing. The approach of this investigation is to leverage the commercial crew landings in the US to obtain measurements as early as possible. While the Moore study used a T-38 X-plane simulation, we will obtain measures during actual T-38 flights. We will also add a lunar landing simulation based on the current concept of operations for the HLS. Further refinements are also proposed in the sensorimotor and cognitive test battery based on the previous study.
Research Impact/Earth Benefits:	Sensorimotor function is critical for spatial orientation, gaze stabilization, and postural stability. This project examines how adaptive changes in sensorimotor and cognitive function may increase the risk of impaired ability to maintain control of vehicles and other complex systems. The goal is to map changes in physiological function with functional measures of manual control. Establishing these relationships will be relevant to how pathophysiological impairments in sensorimotor processing may affect other vehicular control tasks, such as driving with vestibular patients. This study will also be relevant to the development of strategies of how best to allow individuals with impairments perform a manual control override of autonomous driving vehicles. As Shuttle missions were extended, the on-orbit training countermeasure tool became extremely useful for commanders and pilots to practice the landing task sequence to maintain task proficiency. The ability to retain and/or assess manual control efficiency via a simple laptop-based simulation will impact our crew override approach for extended exploration missions. This approach can be easily adapted to a wide variety of simulated vehicle designs to provide similar assessments in other operational and civilian

	populations.
Task Progress:	Flight study integration: During this past year the feasibility assessment by the Research Operations and Integration Element was completed and the Select for Flight CR (change request) was reviewed by the NASA Human Research Program (HRP) Control Board on 25 April 2024. Pending approval and completion of hardware integration, operations can begin as soon as USCV-11 (launching September 2025) for the International Space Station (ISS) crew and PAM-6 (launch late 2025/early 2026). As part of the Feasibility Assessment, there was a change in the scope of the ground measurements to accommodate early postflight testing constraints. A portable fixed base simulation will now be used to perform early testing at the rally airports before the crews return to NASA Johnson Space Center (JSC) for the motion base simulation and sensorimotor test battery on R+1. A new tablet-based simulation will be implemented at multiple time points (R+0, 1 and 4 days) to track the recovery of operational multi-tasking during supervisory control of an automated landing. Piloting simulation development: The lunar landing simulation involving piloting using hand controllers has been integrated into the Robotic On-Board Trainer (ROBoT, Ivkovic et al. 2019) and the CKAS motion base similar to that used in the prior study (Moore et al. 2019). There has been ongoing communication to align the simulation, developed by Draper, to be closely aligned with Artemis Human Landing System (HLS) and Flight Operations Directorate (FOD) training simulations that will be utilized for basic skills training using the HLS design reference vehicle. During this simulation, crewmembers will have the responsibility of monitoring the automatic flight trajectory before being tasked to manually take over and control the attitude and rate-of-descent to the nominal or re-designated landing aimpoint during the approach phase. The primary task of the pilot is to control the flight path and attitude of the lunar lander, with workload measured by secondary tasks requiring subject input (Duda et al. 2015), point of regard using eye tracking (Cheung and Hofer 2003) and a modified Bedford Workload scale following each trial (Hainley et al. 2013). In addition to the motion base simulation, a fixed-based simulation has been developed for early testing at the rally airport and has been designed to facilitate early postflight testing. Multi-Attribute Lunar Tablet Battery Task Software: This portable tablet task provides a set of tasks representative of supervisory control of automated lunar landings, including monitoring enhanced camera display to determine when/if a redesignation of the landing aimpoint is required while monitoring other system displays and warnings that require user inputs. Subjects will monitor the simulation of an out-the-window display during lunar approach to determine if they need to execute a divert to an alternate landing site and use a touch screen to select the site that is within the desired landing zone and free of obstacles. Sensorimotor test battery: The dual-tasking has now been embedded in both tablet and piloting tasks. Motion perception accuracy and precision have been integrated with motion-base sim at JSC. Motion sickness and sleepiness will be obtained at all time points. During this past year, the motion perception measures were implemented on the new CKAS motion base and a pilot study was conducted on eighteen healthy subjects who performed direction-recognition vestibular thresholds in roll-tilt and inter-aural translation, perceptual roll-tilt motion tracking, and a manual control nulling task. This study also supported a master's student in Bioengineering (A. Bollinger) who investigated the optimization of noisy galvanic vestibular stimulation for enhancing vestibular perception and manual control performance. Using a time-optimized approach with a three-down, one-up staircase over 60 trials, the thresholds of the motion perception paradigm yielded results for the baseline none condition in the ranges of previously published literature for inter-aural translation and roll-tilt (Galvan-Garza et al. 2018; Keywan et al. 2019). For the perceptual motion tracking, we implemented a roll-tile motion tracking sum-of-sines over a frequency range of 0.18 – 0.664 Hz (Rosenberg et al. 2018). Both types of perceptual tracking tasks are proposed to help examine how changes in vestibular function relate to proficiency during supervisory and manual control landings (Aim 2).
Bibliography Type:	Description: (Last Updated: 02/20/2026)
Abstracts for Journals and Proceedings	Bollinger AM, Duda KR, Wheelock DH, Moore ST, Wood SJ. "Manual crew override of vehicle landings following G-transitions." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024